INCH-POUND

MIL-DTL-32094
December 19, 2001
SUPERSEDING
NAWC/PD 4.6 1997-26
9 April, 1999

MILITARY DETAILED SPECIFICATION
GLOVE, AIRCREW, COLD/WET PROTECTIVE, HAU-14/P

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of cold/wet protective aircrew glove, HAU-14/P.

1.2 Classification. The gloves will be of the following sizes: 5,6,7,8,9,10 and 11.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Philadelphia, Clothing and Textiles Directorate, Attn: DSCP-COC, Bldg 6-1-D, 700 Robbins Ave, Philadelphia, PA 19111-5096.

AMSC N/A FSC 8315

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this purchase description, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the department of defense index of specifications and standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

A-A-55126	Fastener Tapes, Hook and Pile, Synthetic
A-A-50195	Thread, Aramid
A-A-50199	Thread, Polyester Core, Cotton or Polyester Covered
A-A-59104	Insect Repellent, Clothing and Personal Application
V-T-295	Thread, Nylon
KK-L-254	Leather, Sheepskin, Chrome Tanned

MILITARY

MIL-W-5664	Webbing, Textile, Elastic
MIL-DTL-32075	Label: For Clothing, Equipage, and Tentage (General Use)
MIL-B-44346	Batting, Synthetic Fiber, Medium Density
MIL-C-81393	Cloth, Knitted, Polyamide, High Temperature Resistant, Simplex, Jersey
MIL-C-85636	Cloth, Knit, Jersey, Aramid

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STANDARDS

FEDERAL

FED-STD-191 Textile Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS

BUREAU OF ALCOHOL, TOBACCO AND FIREARMS, DEPARTMENT OF THE TREASURY

Formulas for Denatured Alcohol (27 CFR Part 21)

(Applications for copies of referenced documents should be addressed to the Bureau of Alcohol, Tobacco and Firearms, Department Of The Treasury, 1200 Pennsylvania Ave. Washington, DC 20226.)

ENVIRONMENTAL PROTECTION AGENCY

Regulations for the Enforcement of the Federal Insecticide, Fungicide and Rodenticide Act (40 CFR Part 162)

(Applications for copies of referenced documents should be addressed to the Environmental Protection Agency, 401 M Street SW, Washington, DC 20460.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation.

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AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

AATCC Chromatic Transference Scale
AATCC 127 Water Resistance: Hydrostatic Pressure Test

(Applications for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

ANSI/ASQC Z1.4 Sampling Procedures and Tables for Inspection by Attributes

(Applications for copies should be addressed to the American Society for Quality Control, 611 E. Wisconsin Ave., Milwaukee, Wisconsin, 53203.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-751	Standard Test Methods for Coated Fabrics
ASTM-D-1117	Methods of Testing Nonwoven Fabrics
ASTM-D-1388	Stiffness of Fabrics
ASTM-D-1777	Measuring Thickness of Textile Materials
ASTM-D-1814	Measuring Thickness of Leather Units
ASTM-D-2724	Bonded, Fused, and Laminated Apparel Fabrics
ASTM-D-3776	Mass Per Unit Area (Weight) of Woven Fabrics
ASTM-D-3886	Abrasion Resistance of Textile Fabrics (Inflated Diaphragm Method)
ASTM-D-3787	Bursting Strength of Knitted Goods - Constant-Rate-of Traverse (CRT) - Ball Burst Test
ASTM-D-3887	Tolerances for Knitted Fabrics
ASTM-D-4705	Stitch Tear Strength of Leather, Double Hole
ASTM-D-5035	Breaking Force and Elongation of Textile Fabrics (Strip Test)
ASTM-D-5053	Colorfastness of Crocking Leather
ASTM-D-6193	Seams and Stitches
ASTM-E-96	Water Vapor Transmission of Materials

(Applications for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First articles. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials and components.

3.3.1 Leather, shell. The leather for the finger palm, hand palm, palm patch, fingertips, welt, and pocket pull tab shall be drum dyed black and shall conform to Type A, Class 2, Subclass B, of KK-L-254. The color shall be uniform throughout the grain surface. The minimum wet and dry crocking colorfastness (resistance to rubbing) shall be AATCC Chromatic Transference Scale Grade 2 (wet) and Grade 3 (dry). The thickness shall be not less than 1-1/4 ounces or more than 2 –1/2 ounces at any place in the finished glove. The leather shall have a stitch tear strength of not less than 12 pounds when tested as specified in 4.4.1. The requirements for total ash, chromic oxide, and pH are not applicable. Only tap water shall be used for dampening the leather prior to cutting.

3.3.2 Cloth, shell. The knitted cloth for the remainder of the shell shall conform to MIL-C-81393. The color of the cloth shall be sage green, USAF Shade No. 1565 (see 4.4.1).

3.3.3 Elastic, shell. The elastic for the wrist and back of the hand take-ups shall be 1/2 ±1/32 inches braided cotton elastic having a 50 gauge white elastomer (rubber) and containing 65 picks ±10% and having a hand elongation of 140% ± 10%. The lengths of elastic to be used for each size glove shall be as listed in Table I. The same length of elastic shall be used on the dorsal side (back) of the hand across the wrist and across the palm (see 4.4.1).

TABLE I. Length of elastic take-ups, in inches

Glove size	Palmar side at wrist (across palm/thumb back assembly)	Dorsal side at palm and wrist
5	4	3-5/8
6	4-1/8	3-3/4
7	4-1/4	4
8	4-1/4	4
9	4-5/8	4-1/8

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Glove size	Palmar side at wrist (across palm/thumb back assembly)	Dorsal side at palm and wrist
10	4-5/8	4-1/8
11	4-7/8	4-1/4
Tolerance	+ 1/8 – 0	+ 1/8 – 0

3.3.4 Waterproof insert. The waterproof insert shall be constructed from a waterproof, moisture vapor permeable, three layer laminate. The three-layer laminate shall consist of a textile layer, a film layer and a textile layer. The film layer shall be a microporous polytetrafluoroethylene. The weight shall be 0.5 ± 0.02 ounces per square yard. The film shall be partially impregnated with an oleophobic (oil hating) polymer. The insert shall be properly sized to fit around the glove insulation. Tabs shall be provided on the periphery of the glove pattern for the purpose of securely stitching the insert to the insulation or glove shell. The finished insert shall meet the requirements specified in Table II (see 6.6.1).

TABLE II. Requirements for finished waterproof insert

Characteristic	Requirement
Whole Glove Leak Test @ 4 psi air pressure	Pass
Moisture Vapor Transmission Rate Method B (inside to water) Method BW (inside to water)	 600 g/m^2/day minimum 5000 g/m^2/day minimum
Trapezoidal Tear Warp Fill	 5 lbs. minimum 5 lbs. minimum
Cut Strip Break Warp Fill	 10 lbs. minimum 10 lbs. minimum
Water Permeability after Universal Wear Abrasion Test	No leakage
Mullen Hydrostatic Resistance after DET (diethyltoluamide insect repellent reagent) DET to outside Outside down with taffeta	 160 psi 160 psi

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3.3.5 Lining. The lining shall consist of two types of materials and an auxiliary heating system that attaches to the palmar side of the glove lining. Lining material A is a polyester batting laminate used on the dorsal side of the hand and lining material B is a lightweight, high temperature resistant tricot used on the palmar side. A two-sided pocket closed with hook and pile fastener is located at the palmar side of the wrist. The auxiliary heating system consists of a heat distribution system in the form of a heat transfer bladder (HTB), and a heat source. The HTB is a flexible, liquid containing, hand shaped bladder attached to the palm side of the lining using a pressure sensitive adhesive. The heat source is a commercial, chemical heat pack, which is placed in the pocket of the lining, adjacent to the HTB. The system operates by allowing the liquid in the HTB to pool in the wrist area where it is heated by the heat pack. Subsequent hand movement causes the heated liquid to move freely throughout the HTB, distributing heat to all portions of the hand.

3.3.5.1 Lining material A: backing and facing fabrics. The backing fabric shall be 2-bar tricot knitted with 15-denier nylon monofilament fiber. The facing fabric shall be 100% cotton jersey knit. The fabrics shall meet the physical requirements specified in Table III when tested as specified in 4.4.1.

TABLE III. Physical requirements of Lining material A: backing and facing fabrics

Characteristic	Requirement	
	Backing	Facing
Color	Natural	White
Weight, oz/yd^2:		
min	0.4	3.2
max	0.8	3.5
Fabric count:		
wales/inch (min)	20	30
courses/inch (min)	20	40
Bursting strength, lbs. (min)	20	49

3.3.5.2 Lining material A: insulation. The insulation to be adhered to the backing and facing fabrics shall be a polyester batting meeting the requirements of MIL-B-44346, except that the thickness and weight shall be as specified in Table IV when tested as specified in 4.4.1.

TABLE IV. Physical requirements of Lining material A: insulation

Characteristic	Requirement	
	Minimum	Maximum
Thickness, inches	0.25	-
Weight, oz/yd^2:	2.7	3.3

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3.3.5.3 Finished lining material A: laminate. The finished laminate (nylon backing/polyester batting/cotton facing) for lining material A shall be constructed by bonding the technical back of the nylon backing fabric to one side of the polyester batting and bonding the technical back of the white cotton facing fabric to the other side of the polyester batting. The finished lining laminate shall meet the requirements of Table V (see 4.4.1 and 6.6.2)

TABLE V. Physical requirements of finished lining material A:laminate

Characteristic	Requirement	
	Minimum	Maximum
Thickness, inches Lin A	0.18	0.24
Drape stiffness, inches Lin A	- -	3.53
Bond strength, lbs./3in. width Insulation to Facing, Lin A Insulation to Backing, Lin A	 0.96 0.786	 1.3 1.064

3.3.5.4 Lining material B: Tricot knit for palmar side. The fabric for the palmar side of the glove lining shall be MIL-C-85636 or equal. Prior to the use of the "or equal" product, the contractor shall furnish samples, with supporting data, to the contracting officer for subsequent evaluation by the Naval Air Warfare Center (see 4.4.1 and 6.6.3).

3.3.6 Pocket material. The material for the pocket which holds the chemical heat pack shall be a plain weave having a minimum of 70 ends and a minimum of 50 picks made of 100% cotton or a polyester and cotton blend containing no more than 65% polyester, with a minimum weight of 3.0 oz/yd^2 a maximum weight of 4.1 oz/yd^2. The color shall be white or natural (see 4.4.1).

3.3.7 Fastener tape, hook and pile. The hook and pile fastener tape for the pocket closure shall conform to Type I, Class 1 of A-A-55126A. The color of the fastener tape shall be white. The width of the fastener strips shall be 1/4 inch and the length shall be as listed in Table VI.

TABLE VI. Length of hook and pile fastener strips, in inches

Glove size	Fastener strip length ± 1/8
5, 6 & 7	2-1/4
8 & 9	2-1/2
10 & 11	2-3/4

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3.3.8 Heat distribution component. The heat distribution component shall be 2 layers of 8 mil polyester urethane film dielectrically heat sealed to form a bladder and partially filled with silicone oil, type GE-SF-96-50 (see 6.6.4).

3.3.9 Adhesive. A polyester based thermoplastic adhesive web shall be used for joining the waterproofing insert to the lining at the fingertips. The web adhesive shall be washable and dry cleanable. The adhesive web shall have the properties given in Table VII (see 6.6.6).

TABLE VII. Adhesive Web

Characteristic	Requirement
Weight	0.6 oz/yd^2
Melting point	110-130°C
Minimum activation temperature	100°C

3.3.10 Thread. The thread for stitching the glove shell and hemming shall an aramid filament thread conforming to A-A-50195, Size B, Sage green in color. The thread for stitching the lining shall be natural nylon conforming to Size B, Type I or II, of V-T-295. As an alternate, cotton-covered polyester thread conforming to #50 of A-A-50199 may be used for stitching the lining (see 4.4.1).

3.4 Label/tag. Each item shall be individually bar-coded with a tag for personal clothing items. The tags shall be placed on the outside of the plastic bag, clearly legible and readable by a scanner. The bar coding element shall be a 12 digit national stock number (NSN). The barcode type shall be a medium to high code density and shall be located so that it is completely visible on the item when it is folded and/or packaged as specified. A label, in accordance with MIL-DTL-32075, shall be sewn to the inside of the glove hem. All of the printing shall be legible throughout the expected life of the glove. The label shall include the following information (see 4.4.1):

Cold Wet Protective Flyer's Glove, HAU-14/P
Name of manufacturer
Contract or order No.
NSN
Size

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3.5 Instruction slip. A printed slip of thin paper containing instructions as indicated below shall be inserted in each right hand glove.

Cold/Wet Protective Aircrew Glove Instructions

This glove is intended for use in cold and wet conditions. It contains an auxiliary heating system and a waterproof/breathable insert. The glove shell is made from flame resistant fiber, with sheepskin leather used for the palm.

The auxiliary heating system consists of a heat transfer bladder. This is a hand shaped bladder filled with silicone oil attached to the lining palm, which is heated by a chemical heat pack. Follow the directions on the heat pack package to initiate heating. Place the heat pack in the pocket located at the glove wrist, on the palm side. To assist in warming the fluid in the heat transfer bladder, heat packs should be opened and placed in the heat pack pockets 15 minutes to one half hour before entering a cold environment. Do not remove and lay gloves down in the cold, or store in a cold environment, as the heat transfer fluid will conversely become cold soaked. If the heat packs decrease in warmth, re-expose to air and shake vigorously. Additional heat packs may be obtained using NSN 8415-01-395-3018.

The gloves may be hand washed with mild soap in warm water, not over 120°F. Rinse thoroughly. Wrap in a towel to remove excess moisture. Do not wring or twist. Air dry, without exposure to hot air or full sunlight.

3.6 Design. The gloves shall be a, slip-on, inseamed Flexor® design with a straight thumb, reinforcement patch across the palm and thumb crotch, and elastic take-ups at the wrist and across the back of the hand (see 6.6 and Figure 1). The layers consist of the outer shell made of aramid simplex and leather; a waterproof, breathable insert; and a lining which is both insulated and contains an auxiliary heating system.

3.7 Patterns. Standard patterns shall be furnished to the contractor for use in cutting working patterns (see 6.5). The working patterns shall be identical to the Government patterns. The standard patterns shall not be altered in any way except for the addition of outward notches and are to be used as a guide for making the contractor's working dies. The working dies shall be identical to the standard patterns.

3.7.1 List of pattern parts. The pattern parts, cutting quantities, and fabric for the shell and lining of the gloves shall be as listed in Table VIII.

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TABLE VIII. List of pattern parts

Basic component	Nomenclature	Cutting quantity per pair	Basic material
Shell, Leather	Finger palm (left & right)	2	Leather
	Hand palm (left & right)	2	Leather
	Palm patch (left & right)	2	Leather
	Fingertip thumb (1) (left & right)	2	Leather
	Fingertip index (2) (left & right)	2	Leather
	Fingertip middle (3) (left & right)	2	Leather
	Fingertip ring (4) (left & right)	2	Leather
	Fingertip little (5) (left & right)	2	Leather
Shell, Knit Cloth	Thumb back (left & right)	2	Simplex
	Finger back (left & right)	2	Simplex
	Hand back (left & right)	2	Simplex
Lining, Material A	Finger back (left & right) 1/	2	Batting
	Thumb back (left & right)	2	Batting
	Hand back (left & right)	2	Batting
Lining, Material B	Finger palm (left & right) 1/	2	Tricot
	Hand palm (left & right)	2	Tricot
Heat Pack Pocket	Pocket outer (left & right)	2	Muslin
	Pocket inner (left & right)	2	Muslin

1/ The finger back and finger palm for the lining are cut from the same pattern piece.

3.8 Cutting leather. When cutting the leather, excess stretch in the skins shall be removed by staking, toggling or other means. The skins may be dampened with tap water. All pieces shall be die cut. The welts shall be 3/16 ± 1/16 inch wide and of sufficient length that they extend to within 1/8 inch of each side seam in the finished glove.

3.9 Cutting fabric. All fabric pieces (shell and lining) shall be cut in accordance with the patterns. All pieces except for the fingers shall be cut such that the long dimension of the piece is cut in the direction of the fabric warp. The finger patterns shall be cut on the bias.

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3.10 Marking. Cut parts shall be marked, ticketed or bundled to ensure a uniform size throughout the glove. Any method of marking shall be used except metal fastening devices, sew on tickets, adhesive type tickets that leave traces of adhesive on the material after removal of the ticket.

3.11 Stitches, seams, and stitching. Stitch, seam, and stitching types specified in Table IX shall conform to ASTM-D-6193. Unless otherwise specified, all the seams used in the fabrication of the gloves shall be 1/8 inch wide with a tolerance of + 1/16 inch –0 inches with 9-11 stitches per inch when using a Type 301 stitch and 8-10 stitches per inch when using a Type 401 stitch. All stitching of lapped type seams shall be 1/8 inch from the outside edge with a tolerance of + 1/16 inch –0inch. Trim thread ends within 1/8 inch. Seam allowances shall be maintained with seams sewn so that raw edges, twists, pleats, or puckers will not result. All seams shall start and finish evenly. Ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Back stitching shall not be used as a substitute for thread tails.

3.11.1 Type 304 stitching. Ends of 304 stitching shall be caught in the closing seam of the glove. Thread tension shall be maintained so that there will be no loose tension resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the material sewn.

3.11.1.1 Repairs of type 301 and 304 stitching. Repairs of stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching for type 301 stitching and restarting the stitching a minimum of three stitches back of the end of the stitching for type 304 stitching. The ends of the stitching are not required to be backstitched.

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. For type 301 the stitching shall start a minimum of 1/2 inch back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. For type 304 the stitching shall start a minimum of three stitches back of the defective area, and continue a minimum of three stitches beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the material and restitching in the required manner. The ends of the stitching are not required to be backstitched.

3.11.2 Type 401 stitching. Thread tension shall be maintained so that there will be no loose or excessively tight stitching. All repairs shall be made in accordance with 3.8.1.1a and 3.8.1.1b. Repairs to type 401 stitching may be accomplished by use of stitch type 301.

3.11.3 Manufacturing operations requirements. The gloves shall be manufactured in accordance with the operation requirements specified in Table IX. The contractor is not required to follow the exact sequence of operations listed. Any additional basting or holding stitching used to facilitate manufacturing is permissible provided the thread is removed or does not show in the finished gloves.

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TABLE IX. Table of operations

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCH TYPE	STITCH PER INCH	NEEDLE THREAD	BOBBIN/ LOOPER THREAD	COVER THREAD
1.	Construct outer shell						
	a. Fold each finger on the second notch closest to the fingertip. Align the folded edge even with the lower notch forming a pleat not less than 1/4 "(+1/16, -0") in width, folded toward the back of the hand. Tack the pleats 1/16 inch (+ 1/16 – 0) from edge (see B of Figure 2). Start the stitching not less than 1/8 inch before the pleat and extend it at least 1/8 inch beyond the pleat.	301 (reverse double stitch) or bartack	OSe-1	6-8	B	B	
	b. Join the fingertips to the finger back with a lapped seam 1/4 ± 1/16 inch wide, aligning the notches. The leather fingertips shall be lapped over the hand back. Stitch.	301	LSa-1	9-11	B	B	
	c. Open the finger back (star). The inside square forms a straight line. Align the finger back and hand back with the little finger toward the notched side of the hand back (see A of Figure 2). Insert and center the welt between the two pieces. Stitch all three layers together (see A of Figure 1).	301	SSj-1	9-11	B	B	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCH TYPE	STITCH PER INCH	NEEDLE THREAD	BOBBIN/ LOOPER THREAD	COVER THREAD
	d. Open the finger palm (star). The inside square forms a straight line. Lap the finger palm over the hand palm. Align with the notches on the hand palm. Stitch.	301	LSa-1	9-11	B	B	
	e. Join the thumb fingertip to the thumb back with a lapped seam 1/4 ± 1/16 inch wide, aligning the notches. The leather thumb tip shall be lapped over the thumb back. Stitch.	301	LSa-1	9-11	B	B	
	f. Align the thumb assembly with the palm assembly and stitch together.	301	SSa-1	9-11	B	B	
	g. Position palm patch on palm by aligning the bottom corner (opposite the thumb) with the notch on the little finger side of the palm and stitch. Care should be taken to avoid wrinkles when stitching. Turn at thumb side of palm assembly and continue to stitch for about 1 inch along the edge on the thumb side. The peninsula of the palm patch extends down the back side of the thumb.	301	LSa-1	9-11	B	B	
	h. Elastic take-ups (of the lengths indicated in Table I) shall be zig-zag stitched to the inner side of the pieces at mid-palm and wrist on the hand back and at the wrist on the thumb back. Stitch through the elastic and leather or fabric. The fullness of the glove shall be distributed evenly under the relaxed elastic.	304	SSaa-1	9-11	B	B	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCH TYPE	STITCH PER INCH	NEEDLE THREAD	BOBBIN/ LOOPER THREAD	COVER THREAD
	i. Close shell by stitching together back and palm assemblies. Take care to sew to the inboard side of the finger pleat tacks and catch the ends of the threads used in joining the fingers to the hand pieces. The sides of the palm patch, and the elastic webbing and zig-zag stitching shall be fully caught in the closing seam.	301	SSa-1	9-11	B	B	
	j. Turn hem 1/4 ± 1/16 inch with the back sides of the material together and stitch 1/8 ± 1/16 inch from turned edge.	301	EFa-1	9-11	B	B	
2.	Construct lining						
	a. Pieces cut from Lining A: Open the finger back "star" such that the inside square forms a straight line, align it with the hand back, and stitch together, leaving a thread tail on both ends. Take care to align the little finger toward the little finger side of the hand back.	401	SSa-1	8-10	B	B	
	b. Pieces cut from Lining B: Open the finger palm "star" such that the inside square forms a straight line, align it with the hand palm, and stitch together, leaving a thread tail on both ends. Take care to align the little finger toward the little finger side of the hand palm.	401	SSa-1	8-10	B	B	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCH TYPE	STITCH PER INCH	NEEDLE THREAD	BOBBIN/ LOOPER THREAD	COVER THREAD
	c. Heat pack pocket. Center the pocket pull tab on the back (smooth side) of the pile strip (one side of the hook and pile fastener) and align the long, straight edge of the pile strip with the straight edge of the pull tab.	401	SSa-1	8-10	B	B	
	d. Align the top edge (the edge with the cutaway notch) of the outer pocket with the top edge (cuff edge) of the hand palm making certain that the curved edge of the pocket is aligned with the thumb side and the pocket lays on the wrong side (the side which does not face the hand) of the hand palm. Fold the raw edges toward the outside of the hand palm until they line up with the cutaway notch. Center pile strip assembly on the outside of the lining along the folded edge with pocket pull tab protruding beyond the folded edge. Stitch across the entire folded edge through the center of the pile strip, leaving a thread tail on both ends.	401	SSa-1	8-10	B	B	
	e. Position the hook strip (remaining side of the hook and pile fastener) on the inside of the inner pocket along the top edge, parallel to this edge as indicated by the notches on each side of the pocket. Stitch the hook strip to the inner pocket along its entire length, placing the stitching in the center of the hook strip, and leaving a thread tail on both ends.	401	SSa-1	8-10	B	B	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCH TYPE	STITCH PER INCH	NEEDLE THREAD	BOBBIN/ LOOPER THREAD	COVER THREAD
	f. Align the inner and outer pocket pieces and stitch around the perimeter (sides and bottom), to form the pocket, being careful not to catch the palm lining piece in the seam, leaving a thread tail on both ends.	401	SSa-1	8-10	B	B	
	g. Align the thumb back with the hand palm and stitch together, taking care not to catch the pocket in the seam, leaving a thread tail on both ends. Note that the bottom (cuff) edge of the palm piece should be aligned with the notch on the bottom edge of the thumb back, leaving the thumb back piece protruding beyond the palm piece by about 1/2 inch. The top edge of the inner pocket should be aligned with the edge of the thumb back piece, but not stitched to it.	401	SSa-1	8-10	B	B	
	h. Align the back assembly with the palm assembly and stitch together, taking care not to catch the pocket in the seam, leaving a thread tail on both ends. Catch the ends of the threads used in joining the fingers to the hand at the palm.	401	SSa-1	8-10	B	B	
	i. Place two thickness' of fusible adhesive web strips over the back of each fingertip, starting at the fingertip and extending down the finger for about 1 inch. Cut adhesive web 1/4 inch wider on either side of fingertip. Stitch adhesive web to fingertips through all thickness', placing the stitching within the seam allowance of the existing seam.	401	SSa-1	8-10	B	B	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCH TYPE	STITCH PER INCH	NEEDLE THREAD	BOBBIN/ LOOPER THREAD	COVER THREAD
	j. Attach the heat transfer bladder to the lining palm placing the crotch of the fingers on the bladder 1/4 ± 1/16 inch below the crotch of fingers of the lining. The bladder should be centered over each of the fingers, thumb and palm.						
	k. Position the pocket over the top of the heat transfer bladder. Stitch the pocket assembly to the lining for a distance of at least 1 inch on each side of the pocket, beginning at the top of the pocket . Stitch pocket to the lining on at least one side in an area on the lower half of the pocket for at least one inch, within the seam allowance of the lining closing seam. One or both sides of the pocket may also be stitched to the lining along the entire length.	401	SSa-1	8-10	B	B	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCH TYPE	STITCH PER INCH	NEEDLE THREAD	BOBBIN/ LOOPER THREAD	COVER THREAD
3.	Assembly						
	a. Place the waterproof insert over the insulating/heating lining ensuring that fingertips of the lining are fully extended into the fingertips of the insert. Ensure that there is no gap between the lining fingertip and the waterproof insert fingertip. Apply heat (at approximately 280°F for approximately 4 seconds) to each finger back individually. Take care to avoid having a space between the lining and waterproofing insert fingertips and/or damaging the HTB.						
	b. Using at least 5 stitches per finger, stitch the waterproof insert to the shell at one side of each finger and thumb tip by sewing the tab on the side of the insert tips to the side of the shell tip. Take care to place the stitching within the seam allowance of the shell closing seam.						

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCH TYPE	STITCH PER INCH	NEEDLE THREAD	BOBBIN/ LOOPER THREAD	COVER THREAD
	c. Turn the hem of the outer shell 3/8 to ½ inch over the lining and waterproof insert and stitch closed, 1/4 ± 1/16 inch from the folded edge of the hem, catching the label in the seam. The top edge of the label shall be inserted in the seam at center back, on the interior of the glove, with the printed side displayed.	301	SSbc-1	9-11	B	B	
4.	Finishing.						
	a. The gloves shall be laid off and properly shaped on appropriately sized electrical handforms heated to 175° to 200°F.						

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3.12 Figures. Figures 1, 2, 3, and 4 are furnished for information purposes only. If there are any inconsistencies between the specification and the figures, the specification shall govern.

3.13 Finished lining measurements. This in-process measurement shall be taken to assure that the proper tension is being applied when stitching the lining palm and hand back to the fingers. The finished measurements shall be as shown in Table X (see Figure 4 for locations D-I).

3.14 Finished glove measurements (overall). The finished measurements shall be as shown in Table X. Measurements shall be made using a 1/4-inch wide metal measuring tape which has, at the beginning of the tape, a metal tip positioned at a right angle to the tape. The metal tip shall be hooked over and placed against the edge of the glove when taking measurements. Measurements shall be to the nearest 1/8 inch (See Figure 1 for locations B and C) and shall be taken on pairs of gloves.

TABLE X. Finished measurements, overall (in inches)

Size	Length B	Width C	Lining Width D	Lining Finger Length				
				E	F	G	H	I
5	10 1/2	4 11/16	4-7/16	2-1/2	2-7/8	2-11/16	2	2-1/8
6	10 3/4	4 15/16	4-9/16	2-5/8	3	2-3/4	2-1/8	2-3/8
7	11	5 1/16	4-11/16	2-11/16	3-1/8	2-7/8	2-1/4	2-1/2
8	11-1/4	5 3/16	4-13/16	2-15/16	3-5/16	3	2-3/8	2-5/8
9	11-1/2	5 5/16	4-15/16	3	3-7/16	3-1/8	2-7/16	2-3/4
10	11 5/8	5 7/16	5-1/16	3-1/8	3-9/16	3-1/4	2-1/2	2-7/8
11	11 3/4	5 9/16	5-3/16	3-1/4	3-11/16	3-3/8	2-9/16	3
Toler-ance	± 1/2	± 1/4	+/- 1/16	+ 1/16, -0				

B. The glove length shall be measured from the tip of the middle finger to the edge of the glove body, i.e., the folded hem of the simplex or leather. The glove shall be gently stretched lengthwise and laid flat.

C. The glove width shall be gently stretched out and measured on the palm, 1/2 inch from the base of the little finger.

D. The lining shall be laid flat on a table and measurements shall be taken on the back from side seam stitching to side seam stitching at the horizontal seam joining the lining hand back to the fingers.

3.15 Finished glove leakage test. The finished glove shall be tested for leakage to assure that the waterproof insert has not been damaged during the manufacturing process. The gloves shall not leak when tested as specified in 4.4.5.

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3.16 Heat source component. The heat pack pocket shall be large enough to accommodate an oxygen activated, chemical heat pack, 3- 5/8 (+ /4) inches by 2-1/8 (+ 1/4) inches in size (see 6.2, 6.6.5).

3.17 Pairing. The finished gloves shall be paired, right and left of the same size. A package of the heat source component (see 6.6.5) shall not be included unless specified in the solicitation (see 6.2).

3.18 Workmanship. The finished gloves shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).
c. In- process inspection (see 4.4.2)

4.2 First article inspection. When a first article is required (see 3.1), it shall be examined for the requirements of Table XI.

Table XI. First article examinations and tests

Inspection	Requirement	Inspection Method
Component and Material Inspection	3.3.1, 3.3.2, 3.3.3, 3.3.4, 3.3.5.1, 3.3.5.2, 3.3.5.3, 3.3.5.4, 3.3.6, 3.3.7, 3.3.8, 3.3.9, 3.3.10, 3.3.11	4.4.1
Visual Examination	3.4, 3.5, 3.6, 3.11.1, 3.11.2, 3.11.3, 3.12, 3.18, 3.19	4.4.3
Dimensional Examination: in process	3.14	4.4.2
Dimensional Examination: finished glove	3.15, 3.17	4.4.4
Leakage Testing	3.16	4.4.5

4.2.1 First article units. The pairs of gloves selected as sample units from the first article lot shall be thoroughly checked for dimensional and visual conformance to this purchase description. The sample size of the first article lot shall be specified in the contract. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. The first article inspection samples shall be furnished to the Government as directed by the contract (see 6.2).

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4.3 Conformance inspection. Unless otherwise specified, conformance testing shall be performed on production lot articles for the requirements of Table XII. Sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4.

Table XII. Conformance inspection examinations and tests

Inspection	Requirement	Inspection Method
Visual Examination	3.4, 3.5, 3.6, 3.11, 3.11.1, 3.11.2, 3.11.3, 3.12, 3.18, 3.19	4.4.3
Dimensional Examination	3.14, 3.15, 3.17	4.4.4
Leakage Testing	3.16	4.4.5

4.3.1 Certificate of compliance. Where certificates of compliance are submitted in accordance with 4.4.1, they shall contain verifiable actual test and inspection data. The Government reserves the right to inspect and test the cloth or component to verify the validity of the certification.

4.4 Inspection methods.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected for the requirements of Table XIII.

Table: XIII: Component parts

Material	Characteristic	Paragraph	Test Method
Leather	Material identification Colorfastness - Wet Colorfastness - Dry Thickness Stitch Tear Test	3.3.1 3.3.1 3.3.1 3.3.1 3.3.1	1/ D-5053 2/ D-5053 2/ D-1814 2/3/ D-4705 2
Shell Cloth	Material identification Color	3.3.2 3.3.2	1/ 1/
Elastic	Material identification	3.3.3	1/

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Material	Characteristic	Paragraph	Test Method
Waterproof insert	Film weight	3.3.4	1/
	Leak Test	3.3.4	4.4.5
	MVTR		
	Method B	3.3.4	E 96 2/, 4/
	Method BW	3.3.4	E 96 2/, 5/
	Trapezoidal Tear	3.3.4	D 1117 2/, 6/
	Cut Strip Break	3.3.4	D 5035 2/
	Water Permeability after Universal Wear Abrasion Test	3.3.4	D 3886 2/ 127 7/, 8/, 9/
	Water Permeability after Cold Flex Gelbo	3.3.4	127 7/, 10/
	Mullen Hydrostatic Resistance after DET	3.3.4	D 751 2/, 11/
Lining material A: Backing and facing fabrics for	Material identification	3.3.5.1	1/
	Color	3.3.5.1	1/
	Weight	3.3.5.1	D-3776, Option C 2/
	Fabric Count	3.3.5.1	D-3887 2/
	Bursting Strength	3.3.5.1	D-3787 2/
Lining material A: Polyester insulation	Thickness	3.3.5.2	D-1777 2/ 13/
	Weight	3.3.5.2	D-3776 2/ 14/
Lining material A: Finished laminate	Thickness	3.3.5.3	D-1777 2/ 13/
	Drape stiffness	3.3.5.3	D-1388, Option A 2/
	Bond strength	3.3.5.3	D-2724 2/
Lining material B	Material identification	3.3.5.4	1/
Heat pack pocket material	Material identification	3.3.6	1/
	Weight	3.3.6	1/
	Color	3.3.6	visual
Hook and pile fastener tape	Color	3.3.7	Visual
	Material identification		1/
Heat transfer bladder	Style number or equal	3.3.8	1/
Adhesive	Style number or equal	3.3.9	1/
Thread	Material identification	3.3.10	1/
	Color	3.3.10	1/
Label	Material identification	3.4	1/

1/ A Certificate of Compliance is acceptable.
2/ Refers to ASTM
3/ Determination of leather thickness shall be accomplished by measuring a folded sample area and dividing by two.

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4/ The inside of the insert shall face the water. The free stream air velocity shall be 550 ± 50 fpm as measured two inches above the fabric specimen. The air flow shall be measured at least 2 inches from any other surface. The test shall be run for 24 hours and weight measurements shall be taken at only the start and completion of the test. At the start of the 24-hour test period, the air gap between the water surface and the back of the specimen shall be 3/4 ± 1/16 inch. Five specimens shall be tested.

5/ The inside of the insert shall face the water. The free stream air velocity shall be 550 ± 50 fpm as measured 2 inches above the fabric specimen. The air flow shall be measured at least 2 inches from any other surface. The test shall run for 2 hours and weight measurements shall be taken at only the start and completion of the test. Five specimens shall be tested. The specimen shall be sealed in any manner which prevents wicking and/or leaking of water out of the cup.

6/ Use section 14.

7/ Refers to AATCC.

8/ The abrasion test shall be conducted in the multidirectional mode using 0 emery paper as the abradant. A load of 1 lb. shall be applied to the abradant. The test shall be completed at 300 cycles.

9/ The water permeability shall be measured as specified in AATCC 127, except that a fixed hydrostatic head of 50 centimeters shall be held for 5 minutes, the outside of the insert shall contact the water and five specimens shall be tested. The report shall only include measurement of the appearance of the water droplets. Leakage is defined as the appearance of one or more droplets any place within the 4-1/2 inch diameter test area. The test may be performed using any device which tests the same specimen area at the equivalent pressure. In cases of dispute, the apparatus described in Method AATCC 127 shall be used.

10/ One specimen, 8 inches by 12 inches, shall be cut from the sample unit with the eight inch side in the indicated direction (warp or fill as applicable). The specimen shall be mounted and conditioned for one hour at 0°F in the chamber. Then flexed as specified in Method 2017 of Fed Std 101 except the specimen shall not be aged, the short edges shall not be heat sealed or otherwise joined, and the specimen shall be flexed for 1500 cycles. At the end of the flexing cycle, the specimen shall be removed from the test chamber and conditioned prior to testing for water permeability. Two six inch by eight inch specimens shall be cut from the eight inch by twelve inch specimen and tested for water permeability.

11/ The specimen, 8 inches by 8 inches, shall be laid flat, outside up, on a glass plate, 8 inches by 8 inches by ¼ inch. Three drops of the test liquid DET (as described in additional notes) shall be applied to the center of the specimen. A glass plate of the same dimensions shall be placed on the specimen and a four-pound weight placed in the center of the glass plate of the assembly. After 16 hours, remove the specimen and test immediately for hydrostatic resistance.

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12/ The insect repellent reagent, DET, shall be a solution of 75% by weight (min) of diethyltoluamide and the remainder denatured alcohol. The diethyltoluamide component of the solution shall be a technical grade and contain N, N-diethyl-m-toluamide of not less than 95% purity and the remainder shall consist of entirely or a mixture of ortho or para isomers of N, N-diethyltoluamide. The denatured alcohol component of the solution shall be ethanol, U.S.P. 94.9% by volume and denatured in accordance with Code of Federal Regulations 27 CFR 21, Formula 40 (see 2.1). The insect repellent must be registered with the U.S. Environmental Protection Agency in accordance with the Federal Insecticide, Fungicide and Rodenticide Act (FIFRA) 40 CFR Part 162 (see 2.1.3). (For guidance purposes only, DET insect repellent conforming to A-A-59104 has been used successfully as a reagent in testing.)

13/ Measured using a thickness gauge with a 12 by 12-inch pressure foot applying a load of 0.002 lbs/in^2.

14/ Six 12 by 12 inch square specimens shall be tested. The individual and mean weight of the six specimens shall be reported to the nearest 0.1 ounces per square yard.

4.4.2 In-process inspection. Inspection of sub-assemblies shall be made to ascertain that construction details and the lining width and length which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated non-conformance. The gloves shall be examined after the stitching of the elastic to assure the length of the elastic from the edge of the material to the opposite edge of the material conforms to Table I.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in Table XIV. The lot size shall be expressed in units of gloves. The sample unit shall be one glove and selection shall be by pairs. Defects for pairing shall be classified as a single defect.

TABLE XIV. End item visual defects

Examine	Defect	Classification	
		Major	Minor
Pairing	Right and left glove not the same size	101	
Color	Leather portions of shell not black	102	
	Fabric portions of shell not sage green	103	
	Lining color not white		201
	Leather portions of shell not of uniform color		202
	Fabric portions of shell not of uniform color		203
	Thread color not as specified	104	

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Examine	Defect	Classification	
		Major	Minor
	Thread color other than white used for lining construction		204
Shell or lining design	Not as specified (incorrect material, pattern, etc.)	105	
Cleanness	Clearly noticeable spots or stains affecting appearance		205
Leather	Hole (except a pinhole or needle hole), cut, tear or rip	106	
	Three or more unspecified pinholes or needle holes		206
	Not full grain	107	
	Not soft and flexible	108	
	Grain damage or imperfections seriously affecting serviceability or appearance	109	
	Grain damage or imperfections affecting appearance		207
	Deep open scratch	110	
	Deep wrinkles seriously affecting serviceability or appearance	111	
	Wrinkles affecting appearance		208
	Scar or abraded area seriously affecting serviceability or appearance	112	
	Scar or abraded area affecting appearance		209
	Brittle, bony area, or thin spot seriously affecting serviceability	113	

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Examine	Defect	Classification	
		Major	Minor
	Brittle, bony area, or thin spot affecting serviceability, but not seriously		210
	Loose or spongy leather	114	
	Flanky leather seriously affecting serviceability or appearance	115	
	Flanky leather affecting serviceability, but not seriously		211
Lining	Facing delaminated from batting		212
	Any hole, cut, tear, or rip	116	
Construction, and workmanship; general (applicable to all components unless otherwise indicated herein)	Component misplaced, operation omitted, or operation not properly performed	117	
	Improper marking method used	118	
	Lining reversed so that nylon tricot backing is oriented toward hand	119	
	Component missing or not securely affixed	120	
	Mend in leather or fabric (i.e. mend, patches or splices - not applicable to restitched seam repair)	121	
	Needle chews likely to develop into a hole	122	
Seams and Stitching	Open seam in single stitched seam	123	
	Open seam in any row of a double stitched seam	124	
	Open seam not repaired as specified (when applicable)		213

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Examine	Defect	Classification	
		Major	Minor
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or run-off stitches occur.		
	Not specified seam type	125	
	Not specified stitch type	126	
	Row of stitching omitted	127	
	Loose stitch tension resulting in a loosely secured seam	128	
	Tight stitch tension resulting in cutting of leather or fabric or breaking of stitches when normal pull is applied	129	
	Ends of stitching not secured as specified		214
	Gage of stitching irregular or not as specified, affecting appearance	130	
	Part caught in unrelated row of stitching	131	
	One or two stitches per inch less than specified		215
	Three or more stitches per inch less than specified	132	
	Leather or fabric torn by needles (elongated holes) greater than 1/2 inch	133	
	Leather or fabric torn by needles (elongated holes) up to 1/2 inch		216
	Not backstitched where required	134	
	Backstitching instead of thread tails		217

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Examine	Defect	Classification	
		Major	Minor
	More than the specified maximum number of stitches per inch damaging materials	135	
	More than the specified maximum number of stitches per inch not damaging materials		218
Assembly Detail	Glove not neatly laid off with wrinkles in shell		219
	Glove not neatly laid off with fingers not completely opened	135	
	No inseam construction where specified	136	
	Welt omitted on back	137	
	Welt is more than 3/8 inch from either seam		220
	Welt not completely caught in finger back/hand back seam		221
	Welt too long and caught in closing seam		222
	Pleats missing from any finger (does not include thumb)	138	
	Pull tab omitted from pocket	139	
	Hook and/or pile fastener omitted from pocket	140	
	Pocket hem, pull tab, hook and pile affixed with more than one row of stitching		223
	Shell more than 1/8 inch longer than lining in any finger when on hand or on flat surface.	141	
	Poorly assembled seriously affecting serviceability (e.g., finger distorted or twisted affecting comfort of wearer or causing seam of finger to come in contact with object)	142	

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Examine	Defect	Classification	
		Major	Minor
	NOTE: Any construction defect that results in a malformed glove such as a twisted finger or thumb resulting in any seam lying in the palmar area of the finger or thumb is a major defect. When a finger or thumb is twisted and the seam is rotated to less than 45 degrees from the vertical and does not lie in the palmar area, it is a minor defect.		
	Poorly assembled affecting serviceability but not seriously (e.g., pleats on palm or thumb back created when attaching palm patch, fabric not distributed evenly under elastic take-ups, wrist elastic misaligned more than 1/4 inch at side seams etc.)		224
	Edge of pleat not fully caught in closing seam less than 1/16 inch		225
	Edge of pleat not fully caught in closing seam more than 1/16 inch	143	
	Thread ends more than 1/8 inch in length		226
	Not constructed with specified number of pieces	144	
	Difference in overall length between front and back of glove is more than 1/2 inch	145	
	Elastic webbing across back of hand and at wrist not fully caught in closing seam.	146	
	Gloves contain lint, loose threads or size markings		227
Label and Instruction Slip	Omitted, incorrect, illegible, or misplaced size and identification label		228

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Examine	Defect	Classification	
		Major	Minor
	Size and identification label not securely sewn in seam		229
Heat pack component (if specified in contract)	Heat pack component not included		230
	Heat pack component wrong size		231

4.4.4 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified in Table X. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of gloves. The sample unit shall be one glove and the selection shall be by pairs.

4.4.5 End item leakage testing. The end items shall be tested for leakage in accordance with 4.4.5.1 and 4.4.5.2. The lot size shall be expressed in units of gloves. The sample unit shall be one glove, and the selection shall be by pairs. Any test failure shall be cause for rejection of the lot.

4.4.5.1 Testing apparatus. The test apparatus shall consist of a 3 inch inside diameter tubular rigid polyvinyl chloride test chamber approximately 12 inches in length. The chamber shall be capped approximately 2 inches from the bottom with the cap having a 1/2-inch diameter plastic air escape tube approximately 3 inches long. The test chamber shall be oriented vertically and placed in a container filled with approximately 1 inch of water such that the escape tube vents to the outside of the chamber, with the end of the tube under the water. The top of the tube shall be beveled at a 45° angle on the inside. The top plug shall consist of polyethylene, approximately 4 inches in height, and 4 inches in diameter at the top tapering to approximately 2 inches in diameter at the bottom. An air passage shall be provided vertically through the plug and connected via nominal 1/2 inch tubing through a gage and air regulator to an oil-free compressed air source. A mechanism shall be provided that is capable of clamping the plug into the end of the tube in a reproducible manner such that a seal is provided (see 6.6.1).

4.4.5.2 Testing procedure. The glove shall be inserted into the test chamber to near the cuff. The plug shall be inserted into the cuff of the glove and clamped such that a seal that holds not less than 4 psi is formed with the beveled edge of the test chamber. Air shall be provided to the chamber to a level of 4 psi for a period of not less than 5 seconds. Appearance of bubbles at 5 seconds constitutes failure.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging agency. These gloves will be used aboard ship and the packaging must be environmentally friendly in accordance with directives concerning the packaging of items in this environment.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The glove covered by this purchase document is intended to be worn by military aircrew requiring handwear which provides protection against cold and wet conditions.

6.2 Acquisition requirements.

a. Title, number, and date of this purchase description, including any amendments.
b. Issue of DoDISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.2.1).
c. Type quantity desired. Heat packs should not be included with the gloves unless otherwise included in the solicitation.
d. Applicable patterns.
e. National Stock Number.
f. Whether first article and conformance inspection is required (see 4.2).
g. Name and address of the first article inspection facility (see 4.2), and the name and address of the Government activity responsible for conducting the first article inspection program.
h. Levels of preservation and packing

6.3 First article. When first article inspection is required (see 3.1), the contracting officer should provide specific guidance to offerors on whether the item(s) should be a first article sample, a first production item, or a standard

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production item from the contractor's current inventory, and on the number of items to be tested, as specified in 4.2. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bid should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Non-Government standards. Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.

6.5 Patterns. For access to patterns, address the acquiring activity issuing the invitation for bids. The lengths of some of the pattern pieces have been altered to accommodate the sizing of the heat transfer bladders and will not appear to be nested. Sizing notches are located on the bottom side of each pattern piece. The following notch/sizing code applies:

Number of Notches	Corresponding Glove Size
1	5
2	6
3	7
4	8
5	9
6	10
7	11

6.5.1 Pattern Numbers. The patterns listed in 3.7.1 have been identified with a NAVAIR drawing number for the purpose of configuration control and tracking of pattern changes. These numbers range from 3527AS180-(1to 17) through 3527AS186-(1 to 17). These numbers do not have any effect on manufacturing.

6.6 Source of supply.

6.6.1 Waterproof insert. Glove inserts meeting the requirements of 3.3.4 and a test apparatus meeting the requirements of 4.4.5.1 are available from:

W.L. Gore and Associates
3 Blue Bell Road
Elkton, MD 29121

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The sizing allocation of insert to glove is as follows:

Waterproof insert		Glove size
Size	Part #	
1	NW 134642	5
3	NW 134367	6 & 7
5	NW 134330	8 & 9
7	NW 134480	10 & 11

6.6.2 Insulated lining fabric. Finished insulated lining fabric A meeting the requirements of 3.3.5 is available from:

International Foam Products, Inc.
771 Commercial Avenue
Carlstadt, NJ 07072

6.6.3 High temperature resistant tricot. Fabric meeting the requirements of 3.3.5.4 is available from:

SSM Industries
211 Ellis Avenue
P.O. Box 602
Spring City, TN 37381

6.6.4 Heat distribution component. Heat transfer bladders meeting the requirements of 3.3.8 may be obtained from:

Sealtech
215 Rocky Mountain Road
Athens, TN 37303

The sizing allocation of heat transfer bladder to glove is:

Heat transfer bladder		Glove size
Size	Part #	
1	J84818000	5 & 6
2	J84825000	7, 8 & 9
3	J84832000	10 & 11

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6.6.5 Heat source component. Chemical heat packs meeting the requirements 3.14 will be provided to the contractor as Government Furnished Equipment (GFM) for the purpose of evaluating the size of the pocket. The GFM will be provided by:

Naval Air Warfare Center
Personal Protective Systems Branch, Code 4.6.3.1
Attn: Protective Clothing Team
Building 2187 Suite 2280
48110 Shaw Road, Unit 5
Patuxent River, MD 20670-1906

6.6.6 Adhesive. Adhesive meeting the requirements of 3.3.9 is available from:

Bostik Inc.
268 3rd Avenue
Winder, GA 30680
800-366-7837

Part number SH427S

6.6.7 Elastic. Elastic meeting the requirements of 3.3.3 is available from:

South Carolina Elastic
201 S. Carolina Elastic Rd
P.0. Box 369
Landrum, SC 29356
864-457-3355

6.7 Patent notice. U.S. Patent number 4654896 covers the Flexor® design of this item as described in 3.6 of this specification. The Government does not have a royalty-free license for this patent. Information regarding licensing for production of gloves meeting this specification may be obtained from:

Rinehart Glove Ltd.
P.O. Box 3400
Montrose, CO 81401

6.8 Subject terms (key word) listing.

Leather
Lining
Waterproof insert

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MILITARY INTERESTS:

Custodians

Navy - NU

PREPARING ACTIVITY:

DLA - CT

Project (8315-0416)

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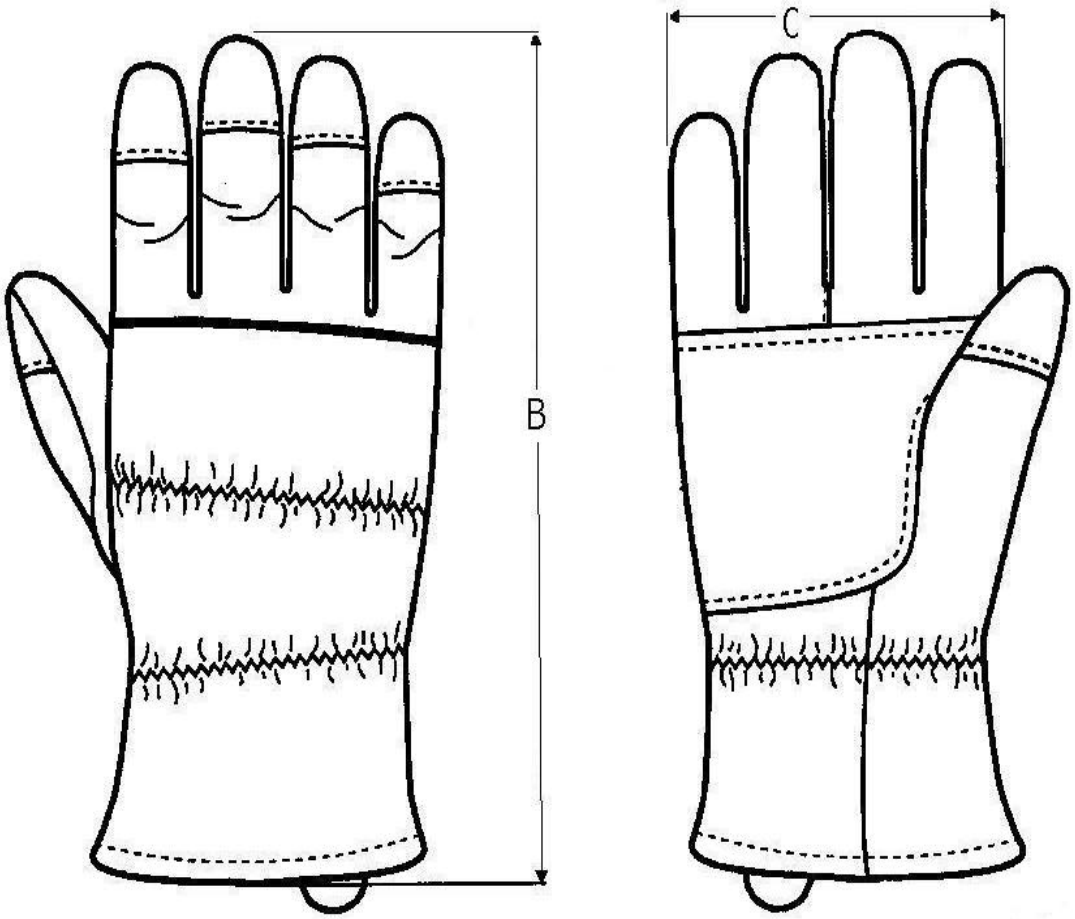


FIGURE 1

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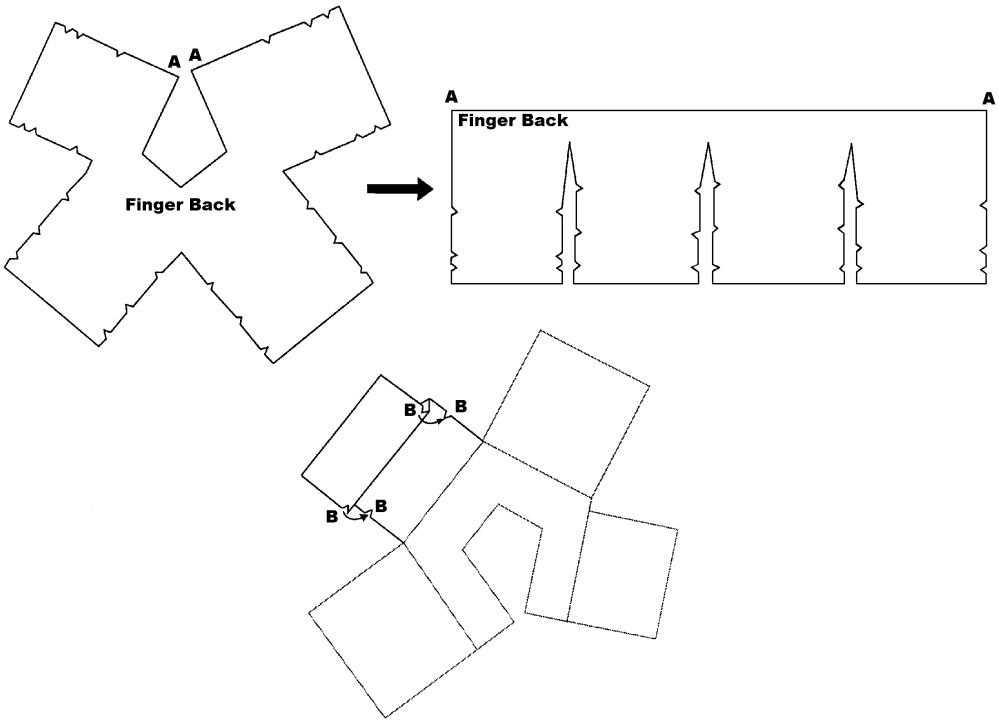


Figure 2

LP41038-03LC1098

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Figure 3: HAU-14/P Palm Patch Position

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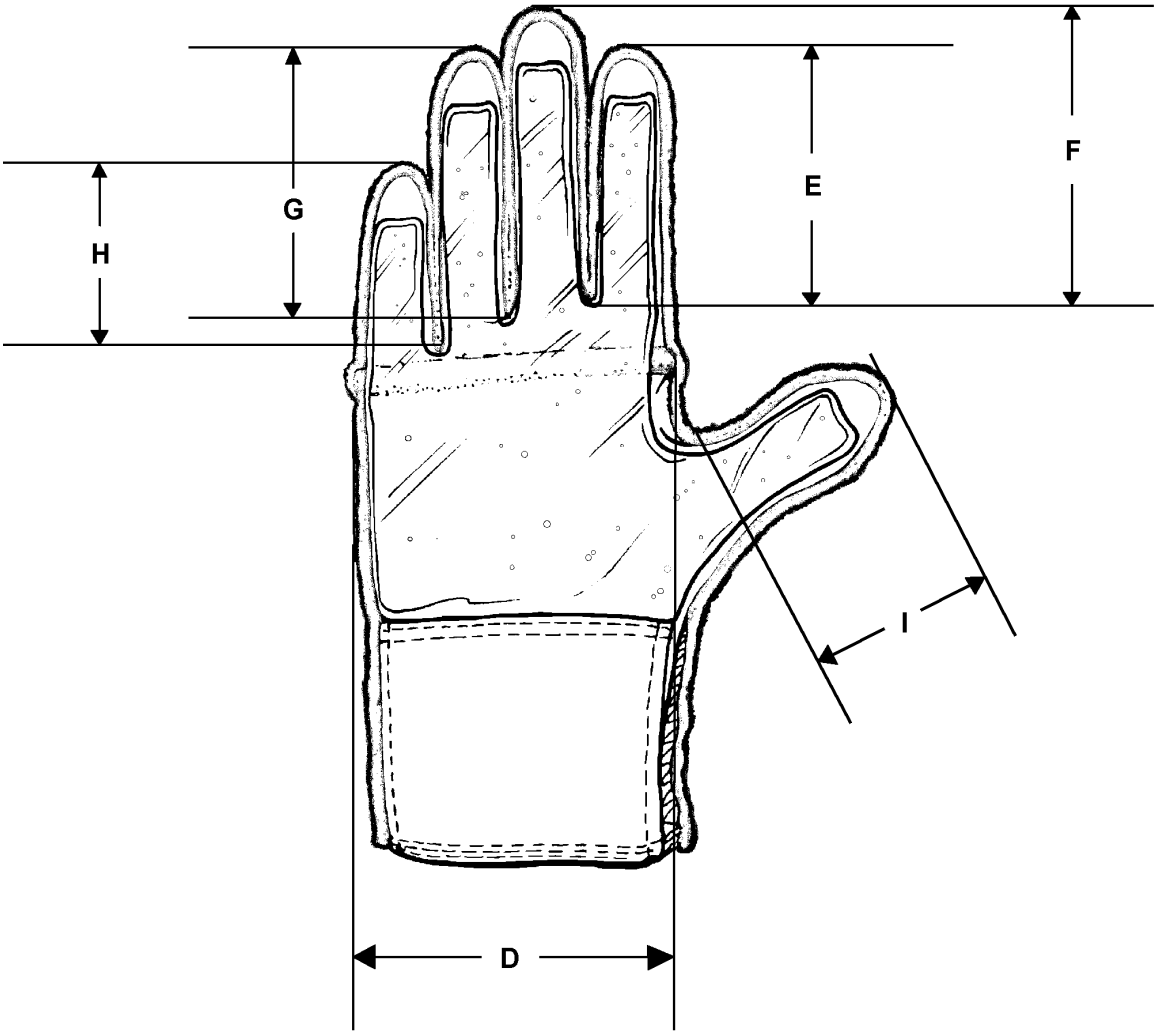


Figure 4: Lining Dimensions

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7 and send to preparing activity.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-32094	**2. DOCUMENT DATE** ***(YYYYMMDD)*** December 19, 2001

3. DOCUMENT TITLE Glove, Aircrew, Cold/Wet Protective, HAU-14/P

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(If applicable)*	7. DATE SUBMITTED *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME DEFENSE SUPPLY CENTER PHILADELPHIA DSCP-C	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN (215) 737- 8035 444-8035
c. ADDRESS *(Include Zip Code)* 700 Robbins Ave (Bldg 6, C&T) PHILADELPHIA, PA 19111-5092	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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